

1. Princeton (von Neumann) vs. Modern Day

Princeton architecture is the original "unified" design.

Similarities

- **Stored-Program Concept:** Both store the program instructions and the data in the same electronic memory.
- **Fetch-Decode-Execute:** Both follow the same fundamental cycle to process every command.
- **Sequential Logic:** Both use a Program Counter (PC) to track and execute instructions one after another by default.

Dissimilarities

- **The Shared Bus:** Princeton uses **one** path for both data and instructions (causing a bottleneck). Modern systems use multiple paths and "caches" to move both at once.
 - **Processing Speed:** Princeton is strictly "one-at-a-time." Modern computers use **Pipelining**, where the CPU starts the next instruction before the current one is even finished.
 - **Single vs. Multi-Core:** Princeton assumes a single processing unit. Modern computers have multiple "cores" (CPUs) on a single chip working in parallel.
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2. Harvard vs. Modern Day

Harvard architecture is the "split" design.

Similarities

- **Dual Pathways:** Both use separate paths for instructions and data to increase speed.
- **Parallel Access:** Both can "read" an instruction and "read/write" data at the exact same clock cycle.
- **Efficiency:** Both are designed to avoid the "bottleneck" that slows down the Princeton model.

Dissimilarities

- **Physical Separation:** In **pure** Harvard, instructions and data are in completely different memory chips. In **Modern Day**, they are only separate inside the CPU (L1 Cache) but share the same RAM.
 - **Address Space:** Harvard has two different "address maps." Modern Day usually has one unified address map for the entire system RAM.
 - **Flexibility:** Pure Harvard is very rigid (hard to change programs). Modern Day is highly flexible, allowing the same memory space to be used for a video game one minute and a word document the next.
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Summary "Cheat Sheet"

- **Princeton:** Everything shares one "pipe" (Simple but slow).
- **Harvard:** Instructions and Data have their own "pipes" (Fast but complex).
- **Modern Day:** A "Hybrid" that uses Harvard's speed inside the chip and Princeton's simplicity for the main RAM.